

Using Partially Observed Confounder Patterns under Monotone Sequential MAR: Efficient Estimation and the Cost of Complete-Case Coarsening

Keivan Bolouri Siahhalehsar

Department of Statistics, University of California, Irvine

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Abstract

Methods for causal inference with several missing confounders often replace the full missingness pattern by a single indicator that all confounders are observed. This complete-case coarsening is convenient, but it discards records in which some confounders are observed and may require a stronger missingness assumption. We study a binary point exposure and two ordered, partially observed baseline confounders under monotone sequential missing at random (MAR). We specialize the standard sequential coarsening-at-random transformation to the full-data efficient influence function for the average treatment effect (ATE), give a tangent-space proof that the resulting observed-data influence function is canonical, and construct a cross-fitted one-step estimator. An exact drift identity yields sequential multiple robustness. In the common sub-model in which both the sequential and complete-case assumptions are valid, we show that the efficiency-bound penalty from complete-case coarsening is

$$\mathbb{E} \left[\frac{1 - \pi_2(Z_0)}{\pi_1(Z_0)\pi_2(Z_0)} \{Q_1(Z_1) - Q_0(Z_0)\}^2 \right] \geq 0.$$

The improvement is strict precisely when the intermediate confounder carries information about the full-data efficient score and second-stage missingness occurs. In an oracle-nuisance experiment with 2,000 samples of size 2,500, retaining the intermediate pattern reduced variance by 4.46% (paired-bootstrap Monte Carlo 95% interval, 2.71%–6.30%), close to the 4.23% reduction obtained by numerical evaluation of the identity. When second-stage observation depended on the intermediate confounder, complete-case MAR failed: the coarsened estimator had bias -0.0322 and 46.9% coverage, whereas the sequential estimator had bias 0.0004 and 95.45% coverage. The results clarify when partial records improve precision and when coarsening instead changes identification.

Keywords: average treatment effect; efficient influence function; missing confounders; monotone missingness; multiple robustness; sequential MAR.

1 Introduction

Observational studies frequently combine two statistical difficulties: treatment is confounded, and one or more baseline confounders are missing. Complete-case analysis is generally inefficient and can be biased. Multiple imputation, inverse-probability weighting, augmented estimators, generalized raking, and targeted learning address parts of this problem, but their validity depends on how the causal and missingness assumptions fit the observed-data structure (Bang and Robins, 2005; Williamson et al., 2012; Seaman and White, 2014; Levis et al., 2025; Benz et al., 2025; Williamson et al., 2026; Wen and McGee, 2026).

Levis et al. (2025) developed robust and efficient ATE estimators under complete-case missing at random (CCMAR). If L_c denotes fully observed confounders, L_p the vector of partially observed confounders, and S the indicator that every component of L_p is observed, their assumption is

$$S \perp\!\!\!\perp L_p \mid (L_c, A, Y).$$

The reduction to S avoids modeling a possibly nonmonotone collection of response patterns. It also means that subjects with only part of L_p observed do not directly contribute their observed confounder values to the final estimating function. Levis and colleagues identify the use of such partial records under alternative assumptions as an important direction for future research.

This paper studies the transparent special case in which the components of L_p have a scientifically meaningful order and are observed monotonically. We write

$$L_c = W, \quad L_p = (L_1, L_2),$$

so the notation L_1, L_2 refines—rather than replaces—the earlier L_c, L_p notation. The possible patterns are: neither partially observed confounder is available; L_1 alone is available; or both are available. Sequential MAR permits the probability of observing L_2 to depend on the already observed L_1 . This model is useful when, for example, an inexpensive screening variable precedes a more burdensome laboratory measure, or when data acquisition follows an ordered protocol.

Sequential inverse-probability augmentation and efficient estimation with monotone missingness are established ideas (Robins et al., 1994; Gill et al., 1997; Tsiatis, 2006; Barnwell and Chaudhuri, 2025). Accordingly, our contribution is not the generic sequential augmentation formula. It is the following causal and comparative analysis:

1. We specialize the monotone sequential construction to the full-data efficient score for a point-exposure ATE with missing confounders, allowing response probabilities to depend on the fully observed treatment and outcome.

2. We give a self-contained observed-likelihood and tangent-space proof of the efficient influence function, followed by an exact remainder identity and a cross-fitted estimator.
3. On the submodel where sequential MAR and CCMAR are simultaneously valid, we derive the exact nonnegative efficiency loss from collapsing the intermediate pattern.
4. We distinguish this efficiency comparison from the case in which second-stage response depends on L_1 . There, CCMAR generally fails, so the comparison concerns identification, bias, and coverage rather than two efficiency bounds for the same model.

The final distinction is practically important. A statement that the pattern-aware method is “more efficient” is justified only under common validity. Outside that submodel, better mean-squared error may result primarily from removing bias under a different, weaker missingness assumption.

2 Observed data, causal target, and assumptions

2.1 Notation and observation patterns

For one individual, let

$$Z_0 = (W, A, Y), \quad Z_1 = (Z_0, L_1), \quad Z_2 = (Z_1, L_2),$$

where W is a vector of fully observed baseline confounders, $A \in \{0, 1\}$ is a fully observed point exposure, Y is a fully observed outcome, and L_1, L_2 are partially observed baseline confounders. Let R_1 indicate that L_1 is observed. Among records with $R_1 = 1$, let C_2 indicate that L_2 is observed, and define $R_2 = R_1 C_2$. Thus

$$(R_1, R_2) \in \{(0, 0), (1, 0), (1, 1)\};$$

the L_2 -only pattern $(0, 1)$ is excluded. The observed datum is

$$O = (Z_0, R_1, R_1 L_1, R_2, R_2 L_2).$$

Table 1: Relationship between the original CCMAR notation and the ordered notation used here.

Object	CCMAR notation	Sequential notation
Always observed confounders	L_c	W
Partially observed confounders	L_p	(L_1, L_2)
Complete-case indicator	S	$R_2 = R_1 C_2$
Intermediate partial record	discarded by coarsening	$R_1 = 1, R_2 = 0$

Let $V = (W, L_1, L_2)$. The causal estimand is

$$\psi = \mathbb{E}\{Y(1) - Y(0)\}.$$

Assumption 1 (Causal identification). For $a \in \{0, 1\}$: (i) consistency, $Y = Y(A)$; (ii) conditional exchangeability, $Y(a) \perp\!\!\!\perp A \mid V$; and (iii) treatment positivity, $\epsilon < e(V) < 1 - \epsilon$ almost surely for some $\epsilon > 0$, where $e(v) = \mathbb{P}(A = 1 \mid V = v)$.

Under [Assumption 1](#), writing $\mu_a(v) = \mathbb{E}(Y \mid A = a, V = v)$,

$$\psi = \mathbb{E}\{\mu_1(V) - \mu_0(V)\}. \quad (1)$$

2.2 Sequential missing at random

Assumption 2 (Monotone sequential MAR). The response process satisfies

$$R_1 \perp\!\!\!\perp (L_1, L_2) \mid Z_0, \quad (2)$$

$$C_2 \perp\!\!\!\perp L_2 \mid (Z_1, R_1 = 1). \quad (3)$$

Let

$$\pi_1(Z_0) = \mathbb{P}(R_1 = 1 \mid Z_0), \quad \pi_2(Z_1) = \mathbb{P}(C_2 = 1 \mid R_1 = 1, Z_1),$$

and suppose $\pi_j \geq \epsilon > 0$ almost surely for $j = 1, 2$.

Treatment and outcome are included in Z_0 , so both response probabilities may depend on A and Y . These are MAR/CAR restrictions because the hazards depend only on information observed at the corresponding stage. They are not outcome-independent missing-not-at-random restrictions.

Proposition 1 (Identification of the full-data law). Under [Assumption 2](#), the full-data density is identified by

$$p(z_0, l_1, l_2) = p(z_0) p(l_1 \mid z_0, R_1 = 1) p(l_2 \mid z_1, R_1 = 1, C_2 = 1). \quad (4)$$

Consequently, under [Assumption 1](#), ψ in [Equation \(1\)](#) is identified from the observed-data law.

Proof. By [Equation \(2\)](#), $p(l_1, l_2 \mid z_0, R_1 = 1) = p(l_1, l_2 \mid z_0)$, hence $p(l_1 \mid z_0, R_1 = 1) = p(l_1 \mid z_0)$. By [Equation \(3\)](#), among $R_1 = 1$, $p(l_2 \mid z_1, C_2 = 1) = p(l_2 \mid z_1)$. Combining these conditional densities with $p(z_0)$ gives [Equation \(4\)](#). The g-formula is a functional of the identified full-data law. $\square$

[Equation \(4\)](#) is a retrospective factorization because A and Y occur in Z_0 . This causes no contradiction with the causal factorization: any identified joint law can also be refactored as $p(v)p(a \mid v)p(y \mid a, v)$, after which [Assumption 1](#) gives the causal interpretation.

3 Efficient influence function under sequential MAR

3.1 Full-data gradient

The efficient influence function for the ATE in the nonparametric full-data model is

$$D_F(Z_2) = G(Z_2) - \psi, \quad (5)$$

where

$$G(Z_2) = \mu_1(V) - \mu_0(V) + \frac{A}{e(V)}\{Y - \mu_1(V)\} - \frac{1 - A}{1 - e(V)}\{Y - \mu_0(V)\}. \quad (6)$$

Define the iterated projections

$$Q_1(Z_1) = \mathbb{E}\{G(Z_2) \mid Z_1\}, \quad Q_0(Z_0) = \mathbb{E}\{G(Z_2) \mid Z_0\}. \quad (7)$$

Theorem 1 (Canonical gradient). *Suppose [Assumptions 1 and 2](#) hold and $D_F \in L_2(P)$. In the non-parametric observed-data model induced by monotone sequential MAR, the efficient influence function for ψ is*

$$D_{\text{seq}}(O) = Q_0 - \psi + \frac{R_1}{\pi_1}(Q_1 - Q_0) + \frac{R_2}{\pi_1\pi_2}(G - Q_1). \quad (8)$$

All functions in a term are evaluated only when the variables required for that term are observed.

Proof. We give the observed-likelihood argument. Write

$$p_0(z_0) = p(z_0), \quad p_1(l_1 \mid z_0) = p(l_1 \mid z_0), \quad p_2(l_2 \mid z_1) = p(l_2 \mid z_1).$$

Under [Assumption 2](#), the observed-data likelihood contribution factorizes as

$$\begin{aligned} p_O(o) &= p_0(z_0)\{1 - \pi_1(z_0)\}^{1-r_1}\{\pi_1(z_0)p_1(l_1 \mid z_0)\}^{r_1} \\ &\quad \times \{1 - \pi_2(z_1)\}^{r_1-r_2}\{\pi_2(z_1)p_2(l_2 \mid z_1)\}^{r_2}. \end{aligned} \quad (9)$$

For regular parametric submodels, the tangent space is the closure of sums of the mutually orthogonal components

$$\begin{aligned} \mathcal{T}_0 &= \{s_0(Z_0) : \mathbb{E}(s_0) = 0\}, \\ \mathcal{T}_1 &= \{R_1 s_1(Z_1) : \mathbb{E}(s_1 \mid Z_0) = 0\}, \\ \mathcal{T}_2 &= \{R_2 s_2(Z_2) : \mathbb{E}(s_2 \mid Z_1) = 0\}, \\ \mathcal{T}_{\pi_1} &= \{(R_1 - \pi_1)h_1(Z_0)\}, \\ \mathcal{T}_{\pi_2} &= \{(R_2 - R_1\pi_2)h_2(Z_1)\}. \end{aligned}$$

Let $q_1 = \mathbb{E}(D_F \mid Z_1) = Q_1 - \psi$ and $q_0 = \mathbb{E}(D_F \mid Z_0) = Q_0 - \psi$. Candidate [Equation \(8\)](#) can be written

$$D_{\text{seq}} = q_0 + \frac{R_1}{\pi_1}(q_1 - q_0) + \frac{R_2}{\pi_1\pi_2}(D_F - q_1). \quad (10)$$

The three terms in [Equation \(10\)](#) belong respectively to $\mathcal{T}_0$, $\mathcal{T}_1$, and $\mathcal{T}_2$: the required conditional means vanish by iterated expectation. Thus D_{seq} belongs to the observed-data tangent space and is orthogonal to the two response-mechanism tangent components.

It remains to verify that D_{seq} represents the pathwise derivative. For $s_0 \in \mathcal{T}_0$,

$$\mathbb{E}(D_{\text{seq}}s_0) = \mathbb{E}(q_0s_0) = \mathbb{E}(D_{\text{F}}s_0).$$

For a full-law score s_1 satisfying $\mathbb{E}(s_1 | Z_0) = 0$, the observed score is R_1s_1 , and conditional expectation over R_1, C_2 gives

$$\mathbb{E}(D_{\text{seq}}R_1s_1) = \mathbb{E}\{(q_1 - q_0)s_1\} = \mathbb{E}(D_{\text{F}}s_1).$$

The last equality uses $\mathbb{E}(s_1 | Z_0) = 0$ and $q_1 = \mathbb{E}(D_{\text{F}} | Z_1)$. Similarly, for s_2 satisfying $\mathbb{E}(s_2 | Z_1) = 0$, the observed score is R_2s_2 and

$$\mathbb{E}(D_{\text{seq}}R_2s_2) = \mathbb{E}\{(D_{\text{F}} - q_1)s_2\} = \mathbb{E}(D_{\text{F}}s_2).$$

Finally, ψ does not depend on π_1 or π_2 , and the inner products of D_{seq} with their score components are zero. Hence Equation (10) represents the derivative along every tangent direction. Because it lies in the tangent space, it is the canonical gradient and has minimum variance among regular influence functions (Tsiatis, 2006). $\square$

Remark 1. The proof is a two-stage specialization of general monotone-MAR/CAR theory (Robins et al., 1994; Gill et al., 1997; Barnwell and Chaudhuri, 2025). Its purpose is to establish rigorously that the causal pseudo-outcome in Equation (6), after sequential projection, yields the efficient observed-data gradient for this particular ATE problem.

4 Estimation, robustness, and inference

4.1 Cross-fitted one-step estimator

Let $\eta = (e, \mu_0, \mu_1, \pi_1, \pi_2, Q_0, Q_1)$. With nuisance estimates trained outside observation i 's validation fold, define

$$\hat{H}_i = \widehat{Q}_{0,i} + \frac{R_{1i}}{\widehat{\pi}_{1,i}}(\widehat{Q}_{1,i} - \widehat{Q}_{0,i}) + \frac{R_{2i}}{\widehat{\pi}_{1,i}\widehat{\pi}_{2,i}}(\widehat{G}_i - \widehat{Q}_{1,i}), \quad (11)$$

and

$$\widehat{\psi}_{\text{seq}} = \frac{1}{n} \sum_{i=1}^n \hat{H}_i. \quad (12)$$

An estimated standard error is

$$\widehat{\text{se}}(\widehat{\psi}_{\text{seq}}) = \left[\frac{1}{n(n-1)} \sum_{i=1}^n (\hat{H}_i - \widehat{\psi}_{\text{seq}})^2 \right]^{1/2}. \quad (13)$$

One practical training sequence is: fit π_1 to all training records; fit π_2 among $R_1 = 1$ records; fit the causal regressions on complete records with inverse-response weights to reconstruct the full law; regress $\widehat{G}$ on Z_1 among complete records to obtain Q_1 ; and regress the stage-two augmented

pseudo-outcome on Z_0 among $R_1 = 1$ records to obtain Q_0 . Other learners are possible, provided they estimate the same nuisance functions and the positivity and convergence conditions below are respected.

4.2 Exact drift and sequential multiple robustness

For arbitrary fixed functions $\bar{G}, \bar{Q}_0, \bar{Q}_1, \bar{\pi}_1, \bar{\pi}_2$, define

$$Q_1^{\bar{G}} = \mathbb{E}(\bar{G} \mid Z_1), \quad Q_0^{\bar{G}} = \mathbb{E}(\bar{G} \mid Z_0),$$

and let $H(\bar{\eta})$ denote Equation (11) with bars in place of hats.

Proposition 2 (Exact missingness-layer drift). *Under Assumption 2,*

$$\begin{aligned} \mathbb{E}\{H(\bar{\eta})\} - \mathbb{E}(\bar{G}) &= \mathbb{E}\left[\left(1 - \frac{\pi_1}{\bar{\pi}_1}\right)(\bar{Q}_0 - Q_0^{\bar{G}})\right] \\ &\quad + \mathbb{E}\left[\frac{\pi_1}{\bar{\pi}_1}\left(1 - \frac{\pi_2}{\bar{\pi}_2}\right)(\bar{Q}_1 - Q_1^{\bar{G}})\right]. \end{aligned} \quad (14)$$

If $\bar{G}$ is formed from $(\bar{e}, \bar{\mu}_0, \bar{\mu}_1)$ as in Equation (6), then

$$\mathbb{E}(\bar{G}) - \psi = \mathbb{E}\left[\frac{\bar{e} - e}{\bar{e}}(\bar{\mu}_1 - \mu_1) + \frac{\bar{e} - e}{1 - \bar{e}}(\bar{\mu}_0 - \mu_0)\right]. \quad (15)$$

Proof. Taking expectation of $H(\bar{\eta})$ over the response indicators conditional on the full data replaces R_1 by π_1 and R_2 by $\pi_1\pi_2$. Subtracting $\mathbb{E}(\bar{G})$ and adding and subtracting $Q_j^{\bar{G}}$ yields Equation (14); the terms involving $Q_j^{\bar{G}}$ reduce to conditional expectations of $\bar{G}$. For Equation (15), condition the barred version of Equation (6) on V , use $\mathbb{E}(A \mid V) = e(V)$ and $\mathbb{E}(Y \mid A = a, V) = \mu_a(V)$, then collect products of the treatment- and outcome-regression errors. $\square$

Corollary 1 (Sequential multiple robustness). *The estimating equation is unbiased for ψ under every nuisance configuration satisfying all three clauses:*

- (i) $\bar{e} = e$ or $(\bar{\mu}_0, \bar{\mu}_1) = (\mu_0, \mu_1)$;
- (ii) $\bar{\pi}_1 = \pi_1$ or $\bar{Q}_0 = Q_0^{\bar{G}}$;
- (iii) $\bar{\pi}_2 = \pi_2$ or $\bar{Q}_1 = Q_1^{\bar{G}}$.

This gives up to $2 \times 2 \times 2 = 8$ sufficient configurations. The result is a layered or sequential form of multiple robustness, not a claim that any arbitrary subset of nuisance models may be misspecified.

4.3 Asymptotic normality

Theorem 2 (Cross-fitted asymptotic linearity). *Use a fixed number of folds. Suppose, uniformly across folds: (i) fitted observation probabilities and treatment probabilities are bounded away from zero (and*

the treatment probability from one); (ii) $\|\hat{H} - H(\eta)\|_2 = o_p(1)$ and the influence functions have uniformly integrable $2 + \delta$ moments for some $\delta > 0$; and (iii)

$$\begin{aligned}\|\hat{e} - e\|_2 \{ \|\hat{\mu}_1 - \mu_1\|_2 + \|\hat{\mu}_0 - \mu_0\|_2 \} &= o_p(n^{-1/2}), \\ \|\hat{\pi}_1 - \pi_1\|_2 \left\| \hat{Q}_0 - Q_0^{\hat{G}} \right\|_2 &= o_p(n^{-1/2}), \\ \|\hat{\pi}_2 - \pi_2\|_2 \left\| \hat{Q}_1 - Q_1^{\hat{G}} \right\|_2 &= o_p(n^{-1/2}).\end{aligned}$$

Then

$$\sqrt{n}(\hat{\psi}_{\text{seq}} - \psi) = \frac{1}{\sqrt{n}} \sum_{i=1}^n D_{\text{seq}}(O_i) + o_p(1) \rightsquigarrow N\{0, \text{Var}(D_{\text{seq}})\}.$$

The variance estimator in Equation (13) is consistent, and the estimator attains the semiparametric efficiency bound.

Proof. Conditional on the training samples, cross-fitting makes each validation-fold nuisance fit fixed. Decompose $\hat{\psi} - \psi$ into the empirical mean of the true influence function, an empirical-process difference on the validation fold, and the conditional drift. The L_2 convergence in (ii) makes the second term $o_p(n^{-1/2})$ without Donsker conditions. Applying Proposition 2 fold by fold and Cauchy–Schwarz bounds the drift by the three products in (iii), up to constants from positivity. Thus the remainder is $o_p(n^{-1/2})$. The central limit theorem applies to the first term. Consistency of the empirical second moment follows from (ii) and uniform integrability. Efficiency follows from Theorem 1. This is the standard cross-fitting argument (Chernozhukov et al., 2018) specialized to Equation (8). $\square$

The conditions do not require every nuisance to converge at $n^{-1/4}$; that rate for each member of a product is merely a simple sufficient condition. Very small response probabilities remain a practical threat even when formal positivity holds, motivating weight diagnostics, truncation sensitivity analyses, and estimators targeted for finite-sample stability.

5 Exact cost of complete-case coarsening

Let $S = R_2$ and suppose the analyst replaces (R_1, R_2) by the single complete-case indicator S . The next restriction creates a submodel in which both approaches identify the same target.

Assumption 3 (Common-validity submodel). *The second-stage response probability does not depend on the intermediate confounder after Z_0 :*

$$\pi_2(Z_1) = \pi_2(Z_0).$$

Then $S \perp\!\!\!\perp (L_1, L_2) \mid Z_0$, with $\mathbb{P}(S = 1 \mid Z_0) = \pi_1(Z_0)\pi_2(Z_0)$, so CCMAR holds. The efficient

influence function based only on $O^{\text{cc}} = (Z_0, S, SL_1, SL_2)$ is

$$D_{\text{cc}}(O^{\text{cc}}) = Q_0 - \psi + \frac{S}{\pi_1 \pi_2} (G - Q_0). \quad (16)$$

Theorem 3 (Efficiency loss from collapsing the pattern). *Under Assumptions 1 to 3,*

$$D_{\text{cc}} - D_{\text{seq}} = \frac{R_1}{\pi_1} \left(\frac{C_2}{\pi_2} - 1 \right) (Q_1 - Q_0), \quad (17)$$

and this difference is orthogonal to D_{seq} . Consequently,

$$\text{Var}(D_{\text{cc}}) - \text{Var}(D_{\text{seq}}) = \mathbb{E} \left[\frac{1 - \pi_2(Z_0)}{\pi_1(Z_0)\pi_2(Z_0)} \{Q_1(Z_1) - Q_0(Z_0)\}^2 \right] \geq 0. \quad (18)$$

Proof. Expanding Equation (16) and subtracting Equation (8) gives Equation (17). Let $\Delta = Q_1 - Q_0$ and $\varepsilon = G - Q_1$. Conditional on Z_1 , $\mathbb{E}(\varepsilon \mid Z_1) = 0$; conditional on Z_1 , $R_1 = 1$, $\mathbb{E}(C_2/\pi_2 - 1) = 0$. Multiplying Equation (17) by the three orthogonal increments in Equation (8) and applying these conditional-mean properties shows $\mathbb{E}\{(D_{\text{cc}} - D_{\text{seq}})D_{\text{seq}}\} = 0$. Finally,

$$\mathbb{E}\{(D_{\text{cc}} - D_{\text{seq}})^2 \mid Z_1\} = \frac{1 - \pi_2(Z_0)}{\pi_1(Z_0)\pi_2(Z_0)} \Delta^2,$$

because R_1 and C_2 are Bernoulli with the stated sequential probabilities. Pythagoras gives Equation (18). $\square$

Corollary 2 (When the gain is strict). *The pattern-aware efficiency bound is strictly smaller if and only if*

$$\mathbb{P}[p_{i2}(Z_0) < 1 \text{ and } Q_1(Z_1) \neq Q_0(Z_0)] > 0.$$

Equivalently, strict improvement requires second-stage missingness and residual predictive information in L_1 about the full-data efficient score after conditioning on Z_0 .

When π_2 depends on L_1 , the complete-case probability conditional on Z_0 averages over an unobserved determinant of selection, and $S \perp\!\!\!\perp (L_1, L_2) \mid Z_0$ generally fails. Then Equation (18) is not an efficiency comparison: the coarsened estimator may target the wrong observed-data functional. This is the distinction between an *information loss* under common validity and an *identification failure* under sequential MAR alone.

6 Simulation studies

6.1 Design

The primary experiment isolates the missingness layer by using oracle nuisance functions. It contains 2,000 independently generated samples of size $n = 2,500$ under each of two response mech-

anisms, with random seed 20260831. No W variables are included. The joint full-data law is specified retrospectively:

$$\begin{aligned} A &\sim \text{Bernoulli}(0.5), \\ Y \mid A = 0 &\sim \text{Beta}(2, 4), \quad Y \mid A = 1 \sim \text{Beta}(4, 2), \\ L_1 \mid A, Y &\sim \text{Bernoulli}\{\text{expit}(-0.6 + 0.5A + 0.25Y + 0.1AY)\}, \\ L_2 \mid A, Y, L_1 &\sim N(A + Y + 2.5L_1Y, 1.25^2). \end{aligned}$$

Numerical refactorization of this joint law gives $e(V)$ and $\mu_a(V)$; numerical integration gives the true ATE $\psi = 0.2558486$. Response at stage one follows

$$\pi_1 = \text{expit}(2 + 0.1A + 0.1Y).$$

In the CCMAR-compatible scenario,

$$\pi_2 = \text{expit}(-1.1 + 0.1A + 0.1Y),$$

whereas in the sequential-MAR scenario,

$$\pi_2 = \text{expit}(-1.9 + 0.1A + 0.1Y + 2.2L_1).$$

The first permits an efficiency comparison; the second violates CCMAR and permits an identification comparison. Confidence intervals use the empirical variance of the oracle influence values in each sample.

Three estimators are reported: the full-data ATE estimator, the coarsened CCMAR estimator in Equation (16), and the pattern-aware sequential estimator in Equation (8). A paired bootstrap over the same Monte Carlo replicates quantifies simulation uncertainty in the variance reduction. Separately, the expectation in Equation (18) was evaluated using 1.5 million numerical integration draws.

6.2 Primary results

The empirical variance ratio (coarsened divided by sequential) was 1.0467 in the common-validity scenario, corresponding to 4.46% lower variance for the sequential estimator. A paired bootstrap with 10,000 resamples gave a Monte Carlo 95% interval of 2.71%–6.30%. Numerical evaluation of Equation (18) produced influence-function variances 0.77466 and 0.74193 for the coarsened and sequential methods, respectively: a ratio of 1.0441 and a 4.23% reduction. The agreement supports the algebraic identity and the simulation implementation.

In the sequential-only scenario, the coarsened estimator's bias was large relative to its standard deviation, and nominal coverage fell to 46.9%. The sequential estimator remained approximately unbiased with 95.45% coverage. This contrast should not be described as an efficiency gain, because

Table 2: Primary oracle-nuisance simulation ($n = 2,500$, 2,000 replicates).

Scenario	Estimator	Bias	Empirical SD	RMSE	95% coverage
CCMAR-compatible	Full data	-0.00029	0.01087	0.01087	94.40%
	Coarsened CCMAR	-0.00049	0.01817	0.01817	94.05%
	Sequential patterns	-0.00047	0.01776	0.01776	94.55%
Sequential-MAR	Full data	0.00043	0.01038	0.01038	95.50%
	Coarsened CCMAR	-0.03222	0.01831	0.03706	46.90%
	Sequential patterns	0.00044	0.01596	0.01597	95.45%

Monte Carlo standard errors for bias ranged from 0.00023 to 0.00041; those for coverage ranged from 0.46 to 1.12 percentage points. The target ATE was 0.2558486.

CCMAR is false in this scenario.

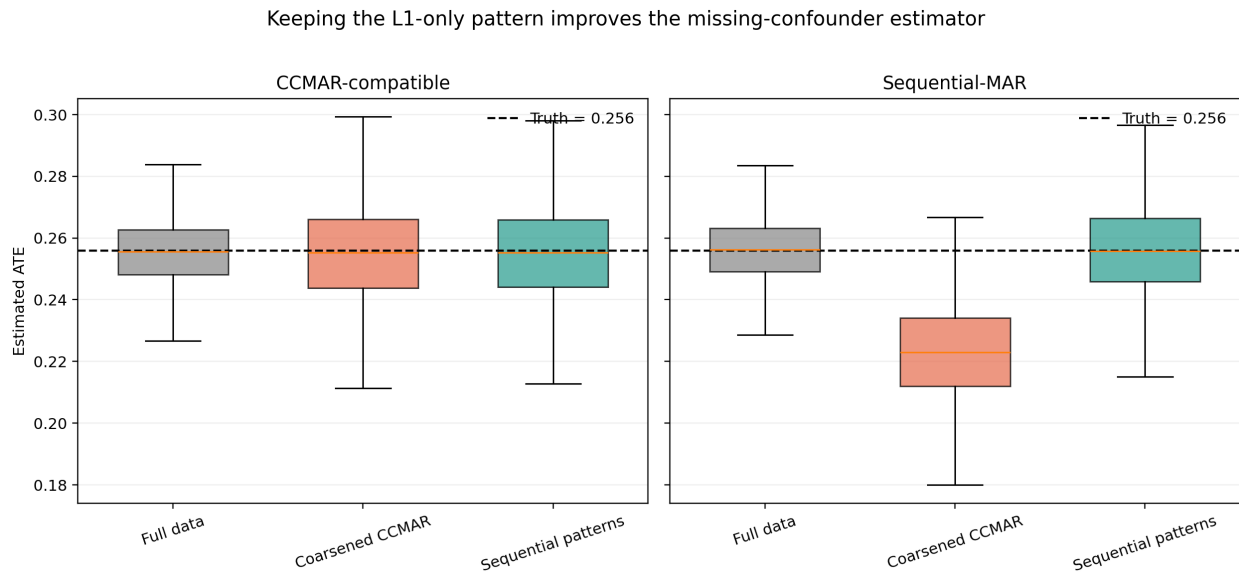


Figure 1: Distributions of oracle estimates across 2,000 replicates. Boxes show the interquartile range and whiskers extend to the conventional $1.5\times$ interquartile range. The dashed line is the true ATE. The CCMAR-compatible panel compares precision; the sequential-MAR panel primarily displays coarsening bias.

The mean pattern proportions were 23.98% complete, 65.09% L_1 -only, and 10.93% neither under common validity; the corresponding values were 31.18%, 57.89%, and 10.93% under sequential MAR.

6.3 Sensitivity analysis

A second oracle experiment used 5,000 replicates per condition. Under common validity, the stage-two intercept was varied to change how many records retained only L_1 . The sequential variance reduction increased from 2.05% with 51.17% complete records to 5.53% with 11.61% complete records (Table 3).

Table 3: Efficiency sensitivity under CCMAR (5,000 replicates per row).

Second-stage missingness	Complete	L_1 only	Sequential variance reduction
Low	51.17%	37.90%	2.05%
Moderate	24.00%	65.06%	3.92%
High	11.61%	77.46%	5.53%

In a separate sequence, intercepts were calibrated to keep the complete-case proportion near 24% while increasing the coefficient γ of L_1 in the second-stage response logit. Coarsened bias and undercoverage worsened monotonically, while sequential bias remained negligible (Table 4).

Table 4: Identification sensitivity as CCMAR departure increases (5,000 replicates per row).

γ	Coarsened bias	Coarsened coverage	Sequential bias	Sequential coverage
0.75	−0.01415	83.30%	−0.00014	95.12%
1.50	−0.02657	61.80%	0.00011	95.12%
2.25	−0.03627	45.74%	0.00011	95.28%

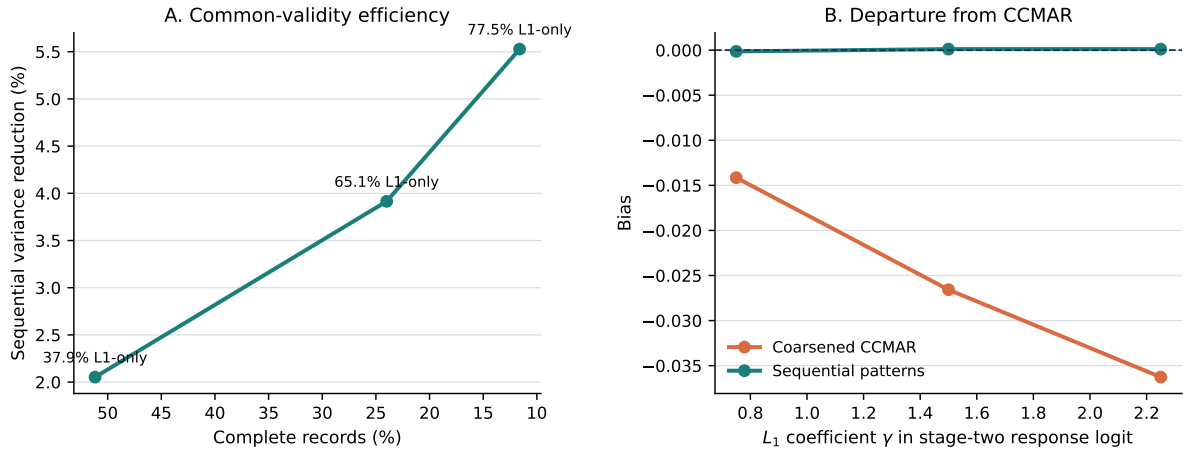


Figure 2: Sensitivity results. Left: efficiency gain under common validity as complete records become rarer. Right: estimator bias as dependence of second-stage response on L_1 increases while the complete-case proportion is held near 24%.

6.4 Estimated-nuisance implementation check

The reproducibility package also contains a fully out-of-fold implementation using weighted treatment and outcome regressions and flexible sequential regressions. In a preliminary 100-replicate check with $n = 4,000$ and true ATE 0.5, sequential-estimator bias was 0.0068 (Monte Carlo standard error 0.0068) under CCMAR and -0.0027 (0.0068) under sequential MAR. Coverage was 91% and 94%, respectively. These results confirm that the end-to-end code runs, but 100 replicates are insufficient for precise evaluation, and the 91% coverage signals that standard-error calibration

or nuisance learning may need improvement. We therefore treat this as an implementation check rather than confirmatory evidence.

7 Discussion

The exact variance identity gives a direct interpretation of partial-record value. The gain is large when three ingredients coincide: many subjects reach stage one but not stage two; stage-two positivity is not too weak; and L_1 predicts the complete-data efficient score beyond Z_0 . Merely having many L_1 -only records is not sufficient if $Q_1 = Q_0$; conversely, an extremely informative L_1 can be of little practical help if almost everyone also has L_2 .

The model comparison also clarifies assumption strength. Under [Assumption 3](#), complete-case coarsening wastes information but remains valid. If L_1 affects the chance that L_2 is observed, sequential MAR may remain plausible because that chance depends on information observed at stage two, whereas CCMAR generally fails after L_1 is hidden inside the all-complete indicator. In that setting, the pattern-aware estimator is not simply a more precise version of the CCMAR estimator; it is based on a different identifying model.

The present results sit between two literatures. General sequential augmentation and efficiency under monotone attrition are well established ([Robins et al., 1994](#); [Gill et al., 1997](#); [Barnwell and Chaudhuri, 2025](#)). Recent causal work addresses complete-case MAR ([Levis et al., 2025](#)), multiple missing confounders or exposures under other MAR/MNAR restrictions ([Wen and McGee, 2026](#)), and broad numerical comparisons of practical missing-confounder estimators ([Benz et al., 2025](#); [Williamson et al., 2026](#)). The narrow value added here is the exact causal score comparison between retaining and collapsing a monotone intermediate pattern. We do not solve the nonmonotone 2^q -pattern projection problem under the original CCMAR model, and we do not claim the generic monotone-MAR influence-function transformation as new.

Several limitations delimit the current evidence. First, all numerical results are synthetic; there is no substantive data application. Second, the main and sensitivity simulations use oracle nuisance functions to isolate the missingness mechanism. Third, the estimated-nuisance study is preliminary and does not yet evaluate a broad library of learners, nuisance misspecification configurations, near-positivity, sample sizes, or confidence-interval corrections. Fourth, monotonicity rules out L_2 -only records and may be implausible in unstructured electronic health records. Fifth, MAR and causal exchangeability are not testable from the observed data alone. Finally, independent expert checking of the canonical-gradient and remainder proofs is necessary before submission.

Before journal submission, the following additions are priorities: (i) a preregistered, larger cross-fitted simulation with at least 1,000–2,000 replicates per design point; (ii) explicit nuisance misspecification and learner comparisons; (iii) weight-truncation and near-positivity sensitivity analyses; (iv) a real or high-fidelity plasmode application with a scientifically defensible ordering of confounder acquisition; (v) a public software interface with unit tests and documented diagnostics; and (vi) independent semiparametric review. Subject to those additions, the paper is best posi-

tioned as a focused methods contribution on the cost of complete-case coarsening rather than as a general solution to arbitrary nonmonotone missing confounders.

Data and code availability

All data analyzed in this manuscript are synthetic. The accompanying reproducibility package contains the Python programs, random seeds, replicate-level CSV files, numerical variance check, generated figures, and the manuscript source. The primary experiment can be reproduced with

```
python code/validate_sequential_mar.py --n 2500 --reps 2000 \
  --seed 20260831 --outdir results/main_verified
```

The original CCMAR implementation of [Levis et al. \(2025\)](#) is available at <https://github.com/alexlevis/flex-ate-confounders-MAR>. That repository is cited as prior software and is not represented as the source of the new sequential-pattern code.

Reproducibility audit

The primary Python simulation, summary bootstrap, and numerical variance identity were rerun in a clean working directory. The regenerated truth, replicate, summary, pattern-proportion, bootstrap-summary, and variance-identity files were byte-identical to the archived outputs. The sensitivity and estimated-nuisance summaries were recomputed from their replicate-level files and matched numerically. A smoke test of the cross-fitted implementation completed successfully. The supplied R mirror was not rerun because R was unavailable in the audit environment; Python outputs are therefore the authoritative results for this manuscript.

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This draft was developed for scientific review. Individuals who contribute substantially to conception, derivation, software, interpretation, or revision should be considered for authorship only after explicit discussion and agreement. No additional author is assigned in this draft.

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A Additional derivation of the efficiency identity

This appendix makes explicit the orthogonality used in [Theorem 3](#). Let

$$\Delta = Q_1 - Q_0, \quad \varepsilon = G - Q_1,$$

so $\mathbb{E}(\Delta \mid Z_0) = 0$ and $\mathbb{E}(\varepsilon \mid Z_1) = 0$. Under [Assumption 3](#),

$$\begin{aligned} D_{\text{seq}} &= Q_0 - \psi + \frac{R_1}{\pi_1} \Delta + \frac{R_1 C_2}{\pi_1 \pi_2} \varepsilon, \\ U &:= D_{\text{cc}} - D_{\text{seq}} = \frac{R_1}{\pi_1} \left(\frac{C_2}{\pi_2} - 1 \right) \Delta. \end{aligned}$$

The product of U with $Q_0 - \psi$ has mean zero by conditioning on Z_1 , R_1 and averaging over C_2 . The product with $R_1 \Delta / \pi_1$ also has mean zero for the same reason. For the last term,

$$\mathbb{E} \left[U \frac{R_1 C_2}{\pi_1 \pi_2} \varepsilon \right] = \mathbb{E} \left[\frac{1 - \pi_2}{\pi_1 \pi_2} \Delta \varepsilon \right] = 0,$$

because the coefficient and Δ are measurable with respect to Z_1 , whereas $\mathbb{E}(\varepsilon \mid Z_1) = 0$. Thus $U \perp D_{\text{seq}}$. Moreover,

$$\begin{aligned} \mathbb{E}(U^2 \mid Z_1) &= \frac{\Delta^2}{\pi_1^2} \mathbb{E} \left[R_1 \left(\frac{C_2}{\pi_2} - 1 \right)^2 \mid Z_1 \right] \\ &= \frac{1 - \pi_2}{\pi_1 \pi_2} \Delta^2, \end{aligned}$$

which establishes the identity.

B Observed-score calculations for response mechanisms

For completeness, consider $S_{\pi_1} = (R_1 - \pi_1)h_1(Z_0)$. Using [Equation \(10\)](#), each inner product $\mathbb{E}(D_{\text{seq}} S_{\pi_1})$ is zero: terms without R_1 vanish because $\mathbb{E}(R_1 - \pi_1 \mid Z_0) = 0$, while terms containing the inverse-weighted residual increments vanish after conditioning successively on Z_0 and Z_1 . For stage two, write the observed score as

$$S_{\pi_2} = (R_2 - R_1 \pi_2)h_2(Z_1) = R_1(C_2 - \pi_2)h_2(Z_1).$$

The q_0 and $q_1 - q_0$ components are orthogonal because the centered Bernoulli residual has conditional mean zero. For the terminal component,

$$\mathbb{E} \left[\frac{R_2}{\pi_1 \pi_2} (D_{\text{F}} - q_1) (R_2 - R_1 \pi_2) h_2(Z_1) \right]$$

is proportional, after response averaging, to $\mathbb{E}\{(D_F - q_1)h_2(Z_1)\} = 0$. Thus the canonical gradient carries no component in either response-mechanism tangent space, as required because the causal target is a functional of the full-data law alone.

C Preliminary estimated-nuisance results

Table 5: Cross-fitted implementation check ($n = 4,000$, 100 replicates; true ATE 0.5).

Scenario	Method	Bias	Empirical SD	RMSE	Coverage
CCMAR-compatible	Full data	0.00165	0.03775	0.03759	96%
	Coarsened	0.00632	0.08966	0.08943	95%
	Sequential	0.00677	0.06757	0.06757	91%
Sequential-MAR	Full data	-0.00111	0.03522	0.03507	96%
	Coarsened	0.00381	0.07828	0.07798	94%
	Sequential	-0.00267	0.06766	0.06737	94%

This study uses a different data-generating process from the oracle experiment and is not intended to reproduce its CCMAR-violation bias. Its purpose is software integration. With only 100 replicates, the Monte Carlo standard error of a 95% coverage estimate is approximately 2.2 percentage points; conclusions about calibration require a substantially larger run.

D Reproducibility inventory

The accompanying archive contains:

- `code/validate_sequential_mar.py`: primary oracle simulation;
- `code/summarize_verified_simulation.py`: Monte Carlo summaries and paired bootstrap;
- `code/verify_variance_identity.py`: numerical evaluation of [Equation \(18\)](#);
- `code/oracle_sensitivity_simulation.py`: efficiency and identification sensitivity designs;
- `code/crossfit_sequential_ate.py`: estimated-nuisance cross-fitted implementation;
- `results/main_verified/`: primary replicate-level results and summaries;
- `results/oracle_sensitivity/`: sensitivity designs, replicates, and summaries;
- `results/crossfit/`: preliminary implementation-check replicates and summary;
- `paper.tex`: manuscript source and `figures/`: generated figures.